

Yash Gupta

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in AWS CloudWatch, System Design, Microservices, building backend, cloud, and embedded systems. Focused on scalable APIs, reliable data pipelines, testing, and measurable production outcomes aligned to the role. Additional strengths include clear cross-functional communication.

TECHNICAL SKILLS

AWS CloudWatch, System Design, Microservices, Python, Golang, C++, SQL, Embedded C, JavaScript, Bash, REST APIs, Data Structures, Algorithms, API Design, AWS EC2, AWS S3, AWS Lambda, AWS RDS, AWS DynamoDB, AWS API Gateway

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Nippon Audiotronix Pvt. Ltd.

Sept 2021 – Present

Gurugram

- Architected and developed backend services for Autonomous Emergency Braking (AEB) system using Python and C++, processing 100+ sensor events/second...
- Implemented real-time sensor fusion algorithms combining LiDAR, camera, and radar data with probabilistic filtering techniques for robust object detection...
- Reverse-engineered and integrated CAN Bus protocols for vehicle telemetry data pipelines, achieving 99.9% message delivery reliability
- Executed ASPIICE-compliant Unit Testing using Unity framework for high-concurrency embedded systems, achieving 95%+ code coverage

PROJECTS

Distributed File Deduplication & Metadata Service on AWS

Nov 2024 - Present

Enterprise-grade hash-based file deduplication service with secure REST APIs and scalable metadata management

Python, AWS Lambda, AWS API Gateway, AWS S3, AWS DynamoDB, AWS IAM, REST APIs, Async/Await

- Designed and implemented hash-based deduplication algorithm reducing storage costs by 60% for duplicate-heavy workloads
- Built serverless REST APIs using API Gateway + Lambda with JWT authentication and role-based access control (RBAC)

CloudMon - Real-Time AWS Resource Monitor Dashboard

Aug 2024 - Oct 2024

Scalable cloud resource usage dashboard with automated cost alerts and ML-based optimization recommendations

Python, Flask, AWS EC2, AWS S3, AWS RDS, Boto3, Docker, PostgreSQL, Jinja2

- Built interactive web dashboard using Flask + Jinja2 displaying real-time EC2, RDS, and S3 resource metrics with live updates
- Implemented Boto3 data fetching pipeline with exponential backoff retry logic for AWS API resilience

Real-Time Data Processing Pipeline for Sensor Telemetry

Mar 2024 - Jul 2024

High-throughput data processing pipeline handling vehicle telemetry with anomaly detection and alerting

Python, Apache Kafka, PostgreSQL, Redis, Unit Testing, CAN Bus

- Architected Kafka consumer application processing 10,000+ messages/second from distributed vehicle sensors
- Implemented anomaly detection using statistical methods and real-time alerting for out-of-threshold sensor readings

EDUCATION

B.Tech in Computer Science & Business Systems | Gyan Ganga Institute of Technology & Sciences

July 2021 – May 2025

Jabalpur | GPA: 8.53/10.0

Coursework: Software Engineering, Operating Systems, Database Management Systems, Computer Networks, Data Structures, Algorithms, Cloud Computing

CERTIFICATIONS

- Oracle Cloud AI Professional – Certified in machine learning and AI on Oracle Cloud
- AWS Cloud Architecting – Specialist in designing scalable AWS infrastructure

ACHIEVEMENTS

- Runner-up, Hackbyte Hackathon 2023 – Developed innovative solution achieving 90%+ accuracy, recognized among 500+ participants
- Shortlisted Round 2, Smart India Hackathon 2023 – Advanced to final round as part of top 1% among 100,000+ teams nationwide